

NORTHWIND SALES DASHBOARD REPORT

1. INTRODUCTION

For this project, I created an interactive Northwind Sales Dashboard using Microsoft Power BI. The main objective of the project was to convert the Northwind dataset into meaningful visual information and analyse sales, orders, quantity sold, freight, customers, products, categories, employees, countries, and monthly performance.

I imported the Northwind dataset into Power BI and used Power Query to clean and transform the data. After preparing the data, I created DAX measures and developed different visualisations such as KPI cards, line charts, column charts, bar charts, tables, slicers, tooltips, and drill-through pages.

The dashboard provides an interactive way to understand the overall sales performance and explore detailed information about products, customers, employees, and orders.

2. OBJECTIVES

The main objectives of this project were:

- To analyse overall sales and order performance.
- To understand total quantity sold and freight.
- To analyse sales performance across product categories.
- To compare order performance across employees.
- To analyse customers across different countries.
- To study monthly sales and order trends.
- To analyse individual product performance.
- To clean and transform the dataset using Power Query.

- To create meaningful DAX measures.
- To use data labels for displaying exact values.
- To use tooltips for providing additional information.
- To create an interactive drill-through page.
- To use slicers for interactive filtering.

3. DATA TRANSFORMATION USING POWER QUERY

After importing the Northwind dataset, I used Power Query Editor to perform data cleaning and transformation before creating the dashboard.

The main data preparation activities included checking the data types of important columns, identifying blank values, removing unnecessary information, and maintaining consistency in product, category, customer, and country-related fields.

The Order Date field was also prepared for monthly and yearly analysis.

Power Query was therefore used as the data preparation stage before the cleaned data was loaded into Power BI.

4. MEASURES USED

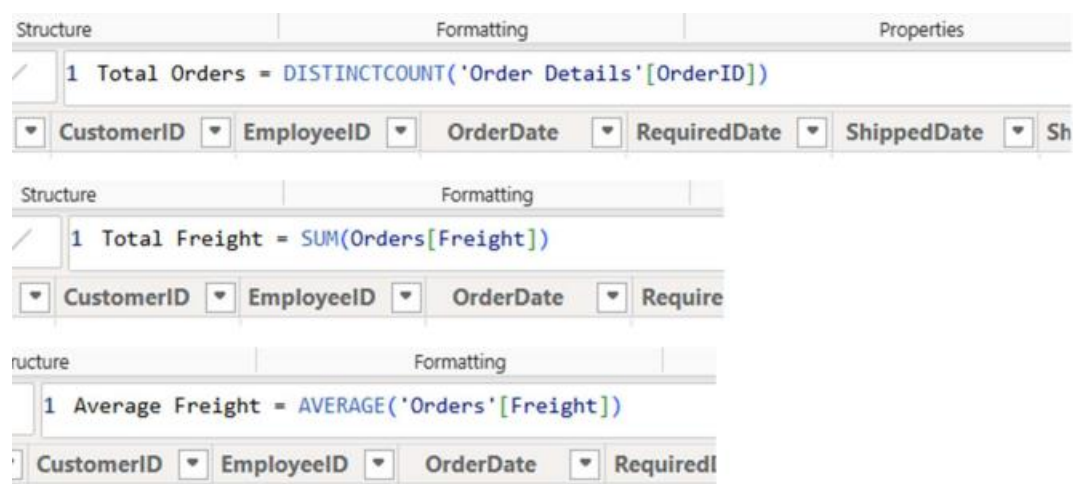
I created different DAX measures to analyse the Northwind dataset.

The major measures used in the dashboard include:

Measure	Purpose
Total Orders	Shows the total number of orders
Total Sales	Shows the total sales generated
Total Quantity Sold	Shows the total quantity of products sold

Measure	Purpose
Total Freight	Shows the total freight value
Average Freight	Shows the average freight value
Average Order Value	Shows the average sales value per order
Number of Customers	Shows the number of customers
Number of Products	Shows the number of products

These measures were used in KPI cards, charts, tables, tooltips, slicers, and detailed analysis.



5. KPI Cards

I created KPI cards to provide a quick summary of the main business figures.

The major KPI cards include:

- ❖ Total Sales
- ❖ Total Orders
- ❖ Total Quantity Sold
- ❖ Total Freight
- ❖ Average Order Value

❖ Number of Customers

These KPI cards provide an overall view of the Northwind business performance before moving into detailed visual analysis.



6. VISUALISATIONS USED

6.1 ORDERS BY MONTH

I created a Line Chart using the Order Date month and Total Orders.

This visual helps me understand how the number of orders changes from month to month. It makes it easier to identify periods with relatively higher or lower order activity.

Data Labels:

Data labels can be used to display the exact number of orders on the chart.

Tooltip:

When I hover over a particular month, the tooltip provides the month and corresponding Total Orders value.

6.2 SALES BY PRODUCT CATEGORY

I created a Column Chart using Category Name and Total Sales.

This visual helps me compare the sales generated by different product categories and understand the contribution of each category to overall sales.

Data Labels:

The data labels display the exact sales value for each category.

Tooltip:

The tooltip provides:

- ✓ Category Name
- ✓ Total Sales
- ✓ Total Quantity Sold

This allows me to obtain additional numerical information without overcrowding the main chart.

6.3 ORDERS BY EMPLOYEE

I created a Bar Chart using Employee Name and Total Orders.

This visual helps me compare the number of orders associated with different employees.

Data Labels:

The exact Total Orders value can be displayed beside each bar.

Tooltip:

When I hover over an employee, the tooltip displays the Employee Name and Total Orders.

6.4 CUSTOMERS BY COUNTRY

I created a visual to analyse the number of customers across different countries.

The Country field is used as the category and Number of Customers is used as the value.

This visual helps me understand the geographical distribution of the Northwind customer base.

Data Labels:

The data labels display the number of customers for each country.

Tooltip:

The tooltip provides:

- ✓ Country
- ✓ Number of Customers
- ✓ Total Sales

6.5 PRODUCT-WISE ANALYSIS

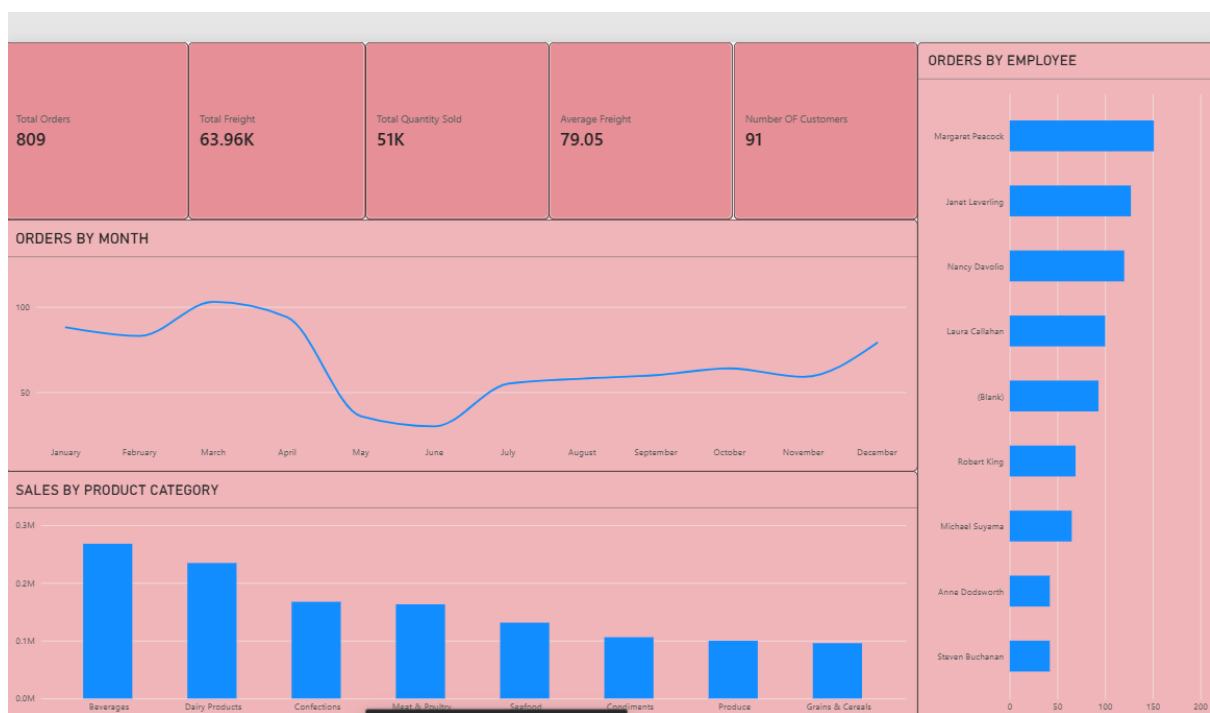
I created a detailed table containing product-level information.

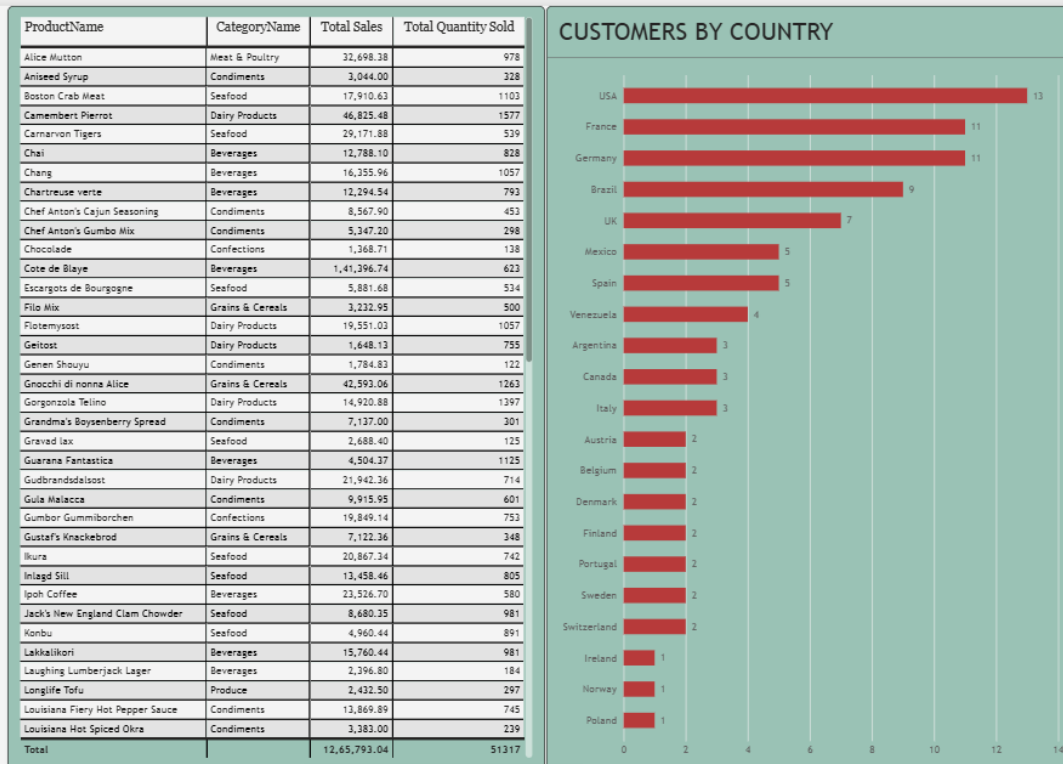
The table includes:

- ✓ Product Name
- ✓ Category Name
- ✓ Total Sales
- ✓ Total Quantity Sold
- ✓ Total Orders

This table allows me to compare individual products and understand their sales and quantity performance.

It also helps identify products that contribute significantly to the overall business performance.





7. CATEGORY NAME STANDARDISATION

Data transformation was performed to maintain consistency in category names.

The category information was checked during the Power Query transformation stage to ensure that the category names were consistent throughout the dataset.

The standardised category information is then used throughout the dashboard.

This helps maintain consistency in:

- ✓ Category slicer
- ✓ Category-related charts
- ✓ Product table
- ✓ Tooltip
- ✓ Category analysis

- ✓ Drill-through analysis

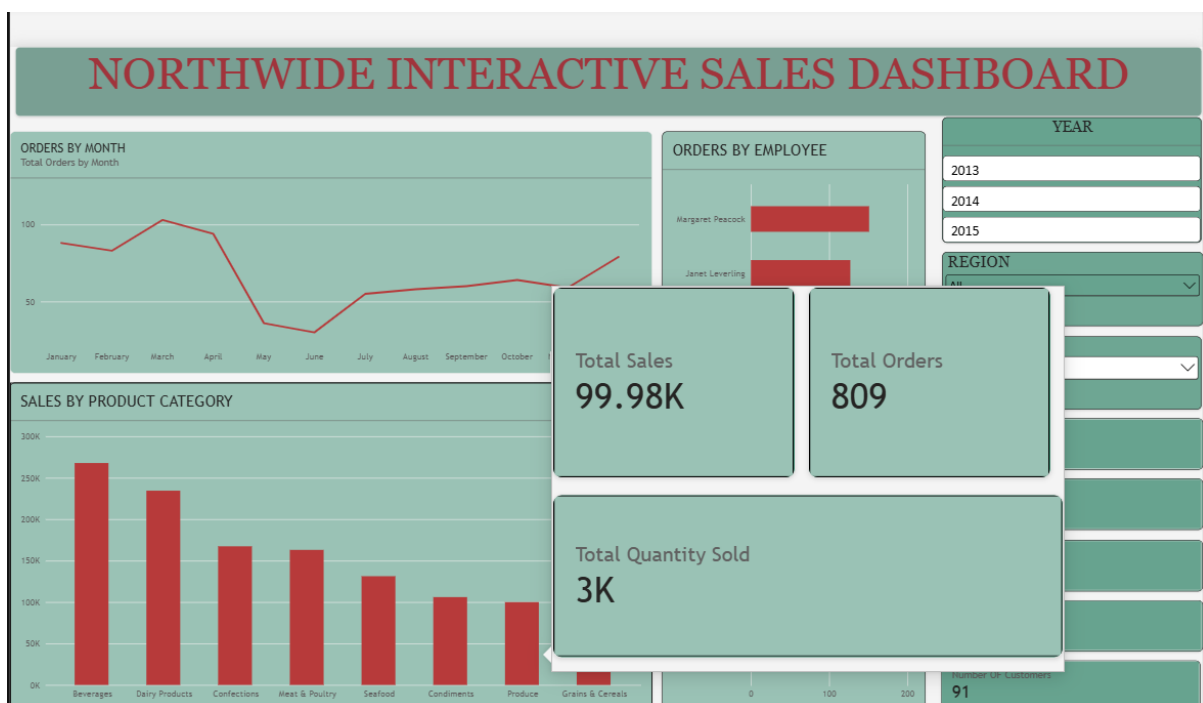
8. TOOLTIPS

I used tooltips to provide additional information when the user moves the cursor over a visual.

The main tooltip information includes:

Visual	Tooltip Information
Orders by Month	Month + Total Orders
Sales by Category	Category Name + Total Sales
Orders by Employee	Employee Name + Total Orders
Customers by Country	Country + Number of Customers
Product Analysis	Product + Sales + Quantity

Tooltips help keep the main dashboard clean while still allowing users to access detailed numerical information.



9. DATA LABELS

I used data labels in selected charts to display exact values directly on the visuals.

For example, data labels can display:

- ✓ Total Sales for each category
- ✓ Total Orders for each employee
- ✓ Number of Customers for each country
- ✓ Total Orders for monthly analysis
- ✓ Total Quantity Sold for selected products

Data labels make it easier to understand the exact numerical values without depending only on the visual axis or tooltip.

10. SLICERS AND INTERACTIVITY

I used slicers to make the dashboard interactive.

The main filtering fields include:

- ✓ Country
- ✓ Year
- ✓ Category
- ✓ Product

When a particular value is selected, the related visuals and KPI cards automatically update based on the selected filter.

This allows the user to analyse the Northwind data from different perspectives and compare specific sections of the business.

11. DRILL-THROUGH ANALYSIS

I also added a Drill-through feature to the Northwind dashboard.

The purpose of drill-through is to move from a summary visual to a separate detailed page based on the selected data point.

For example, when a user selects a particular product or category from the main dashboard and uses the Drill-through option, Power BI can take the user to the detailed analysis page.

The detailed page provides information such as:

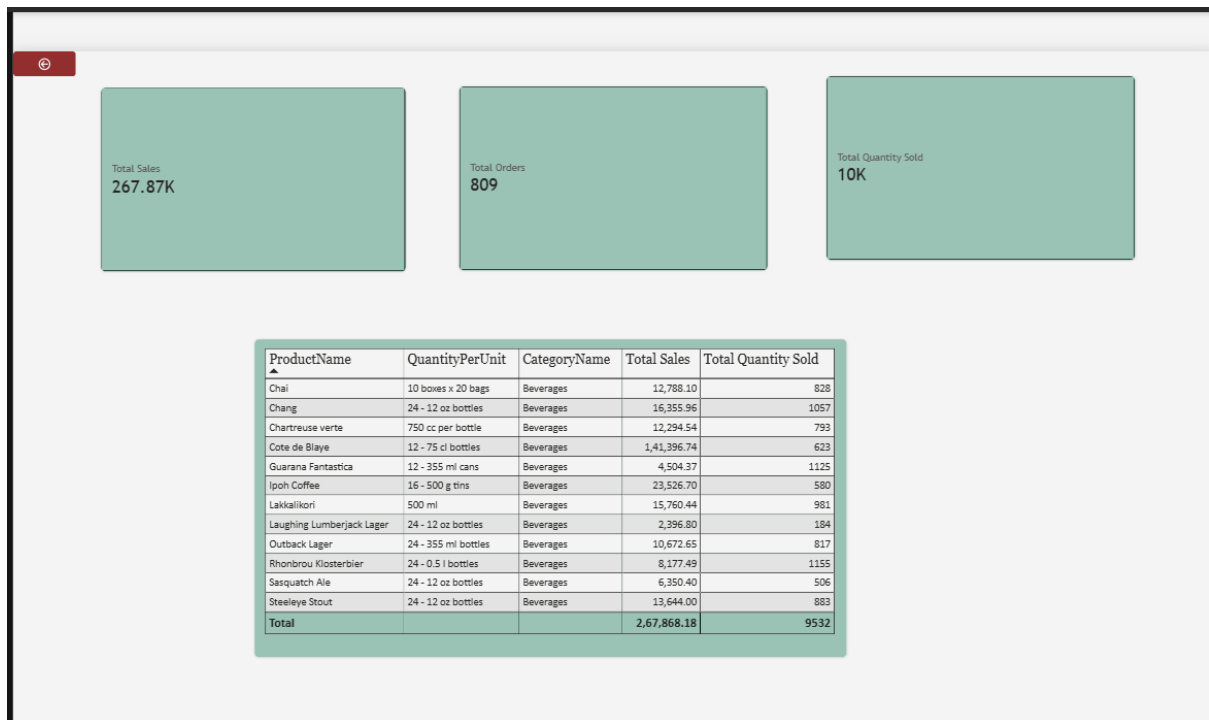
- ✓ Selected Product or Category
- ✓ Total Sales
- ✓ Total Orders
- ✓ Total Quantity Sold
- ✓ Average Order Value
- ✓ Related product information

This feature is useful because the main dashboard provides a high-level summary, while the drill-through page provides deeper information about the selected item.

Benefits of Drill-through

- ✓ Provides detailed analysis without overcrowding the main dashboard.
- ✓ Allows users to investigate a selected product or category.
- ✓ Maintains the selected filter context.
- ✓ Connects summary-level analysis with detailed analysis.
- ✓ Makes the dashboard more interactive.

I also included a Back button on the detailed page so that the user can return to the main dashboard after completing the drill-through analysis.



12. DASHBOARD PAGES

The Northwind Power BI report contains the following pages:

1. SALES OVERVIEW

Provides the main summary of sales and order performance.

2. PRODUCT & CUSTOMER ANALYSIS

Provides detailed analysis of products and customers.

3. INTERACTIVE SALES DASHBOARD

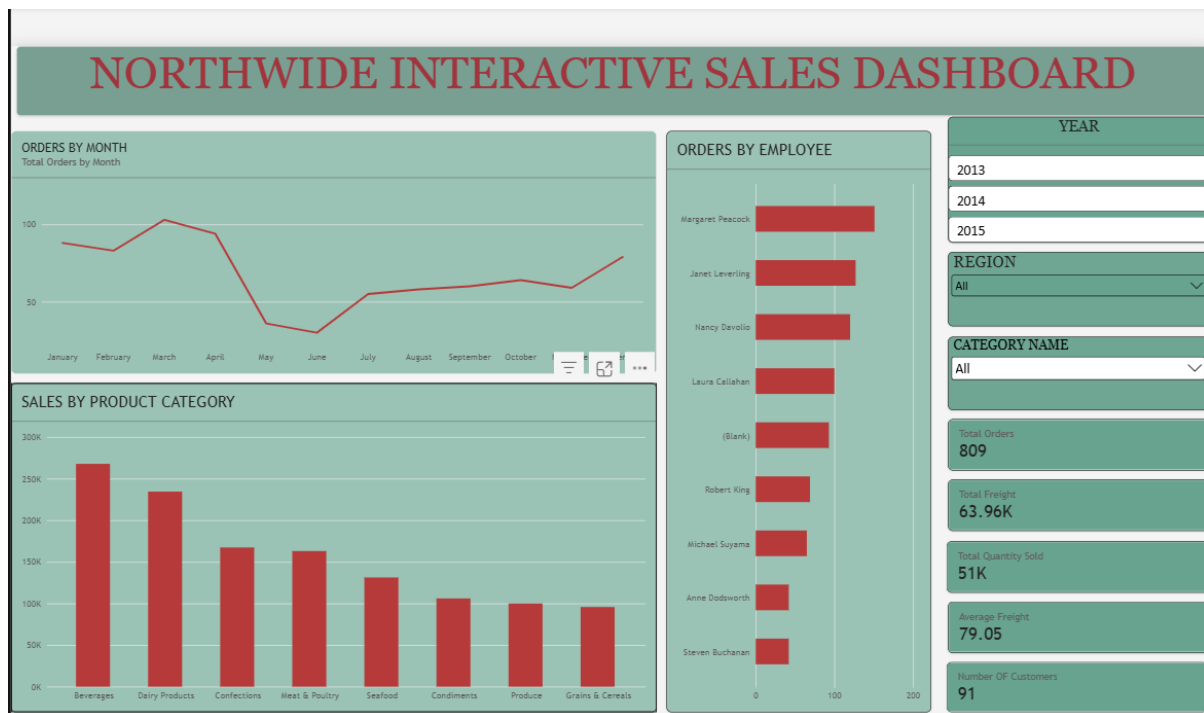
Provides interactive KPI cards, charts, slicers, tooltips, and other analysis.

4. CATEGORY TOOLTIP

Provides additional information related to category-level analysis.

5. CATEGORY DETAILS

Provides detailed information through the drill-through functionality.



13. KEY FINDINGS

From the dashboard, I was able to analyse:

- Overall sales performance using the Total Sales KPI.
- Overall order performance using the Total Orders KPI.
- Total quantity sold across the dataset.
- Total and average freight using the relevant measures.
- Customer distribution across different countries.
- Sales contribution from different product categories.
- Order activity across different employees.
- Monthly changes in sales and order performance.
- Product-level sales and quantity information.
- Category-level information through tooltips.
- Detailed product or category information using the drill-through page.

- The effect of selected country, year, category, and product filters through slicers.
- The importance of Power Query in preparing the dataset before visualisation.
- The usefulness of DAX measures in analysing business performance.

14. CONCLUSION

Through this Northwind Power BI project, I learned how to convert raw business data into an interactive dashboard. I used Power Query to clean and transform the dataset and created DAX measures to analyse sales, orders, freight, quantity, customers, and products.

I used KPI cards, charts, tables, slicers, data labels, and tooltips to present the information clearly. I also implemented drill-through functionality to move from the summary dashboard to a detailed analysis page.

Overall, this project helped me understand how Power Query, DAX, data visualisation, slicers, tooltips, data labels, and drill-through work together in Power BI to create an interactive business analysis dashboard.